



**NARASARAOPETA ENGINEERING COLLEGE (AUTONOMOUS)**  
**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**

**2025-2026**

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| <b>Batch Number</b>          | AB-11                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Team Members</b>          | S. Md. Sakhib (22471A0539)<br>D.C.V. Siva Nagi Reddy (22471A0516)<br>G.Venkata Ganesh (22471A0525)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Guide</b>                 | Dodda Venkat Reddy M. Tech (Ph.D.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Title</b>                 | <b>Climate-Smart Cotton Yield Prediction: A Comparative Study of Random Forest and LightGBM</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Domain/Technology</b>     | MACHINE LEARNING                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Base Paper Link</b>       | <a href="https://ieeexplore.ieee.org/document/10568958">https://ieeexplore.ieee.org/document/10568958</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Dataset Link</b>          | <a href="https://drive.google.com/file/d/1oGjkzjlTE0nCcwVDCqIyJMrAyMyCjVXL/view?usp=drive_link">https://drive.google.com/file/d/1oGjkzjlTE0nCcwVDCqIyJMrAyMyCjVXL/view?usp=drive_link</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Software Requirements</b> | Browser: Any latest browser like Chrome<br>Operating System: Windows 7 Server or later Python                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Hardware Requirements</b> | System Type: Intel Core i5 or above RAM: 8 GB<br>Number of cores:5<br>Number of Threads: 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Abstract</b>              | <p>The United States cotton industry is devoted to sustainable production strategies that reduce water, land, and energy consumption while enhancing soil health and cotton yield. Climate-smart agricultural solutions are being developed to increase yields and reduce operational costs. However, crop yield prediction is challenging because of the complex and nonlinear interactive effects of cultivar, soil type, management, pests and diseases, climate, and weather patterns on crops. To address this challenge, the machine learning (ML) method was used to predict yield, considering climatic change, soil diversity, cultivars, and fertilizer applications. Field data were collected over the southern US cotton belt in the 1980s and the 1990s. A second data source was generated from the process-based cotton model GOSSYM to reflect the most recent effects of climate change over the last six years (2017–2022). We focused on nine locations in three southern states: Texas, Mississippi, and Georgia. The accumulated heat for each set of experimental data was used as an analogue for the time-series weather data to reduce the number of computations. The Random Forest (RF) regressor, Support Vector Regression (SVR), Light Gradient Boosting Machine (LightGBM) regressor, Multiple Linear Regression (MLR), and neural networks were evaluated. Cross-validation was performed to obtain an improved model that did not suffer from overfitting. The RF regressor achieved an accuracy of 97.75%, with an of roughly 0.98 and a root mean square error of 55.05 kg/ha. The</p> |

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|  | results demonstrate how a simple and robust model can be developed and utilized to help cotton climate-smart efforts. |
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**Signature of the student(s)**      **Signature of the Guide**      **Signature of the project coordinator**